

Nonlinear Hall Effect Analysis Report

Extracted Nonlinear Coefficients

Coefficient	Value
α_1	4.898470e-02
α_2	-1.567957e-05
α_3	-4.812896e-05

Goodness of Fit

Metric	α_1	α_2	α_3
R ²	1.00000	0.91462	0.95444
RMSE	4.913e-05	5.186e-06	1.086e-05

Computed Harmonic Response

Component	Voltage (V)
DC	-3.135914e-05
V_{ω}	9.768062e-02
$V_{2\omega}$	-3.135914e-05
$V_{3\omega}$	-9.625791e-05

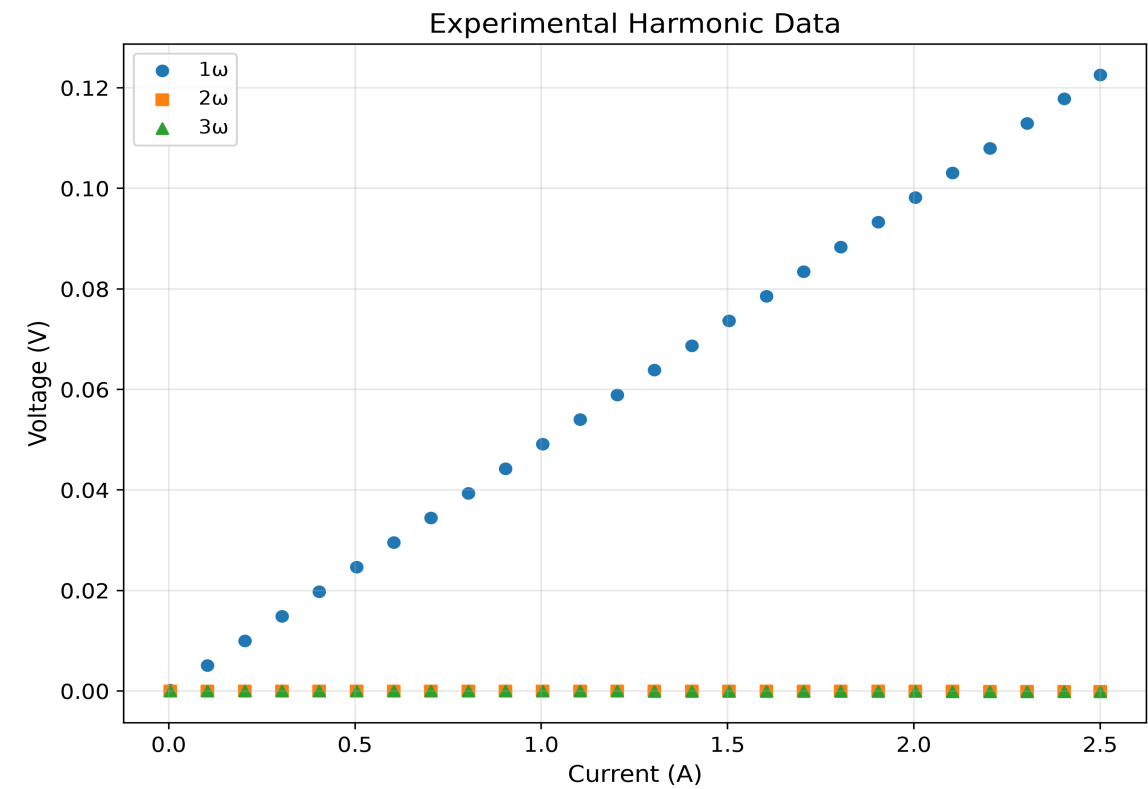
Frequency Mixing Products

Component	Frequency (Hz)	Amplitude
f1	1000.00	1.000000e+00
f2	1200.00	1.000000e+00
f1 + f2	2200.00	1.567957e-05
f1 - f2	200.00	1.567957e-05
2f1	2000.00	1.567957e-05
2f2	2400.00	1.567957e-05
2f1 + f2	3200.00	4.812896e-05
2f1 - f2	800.00	4.812896e-05

$2f_2 + f_1$	3400.00	4.812896e-05
$2f_2 - f_1$	1400.00	4.812896e-05

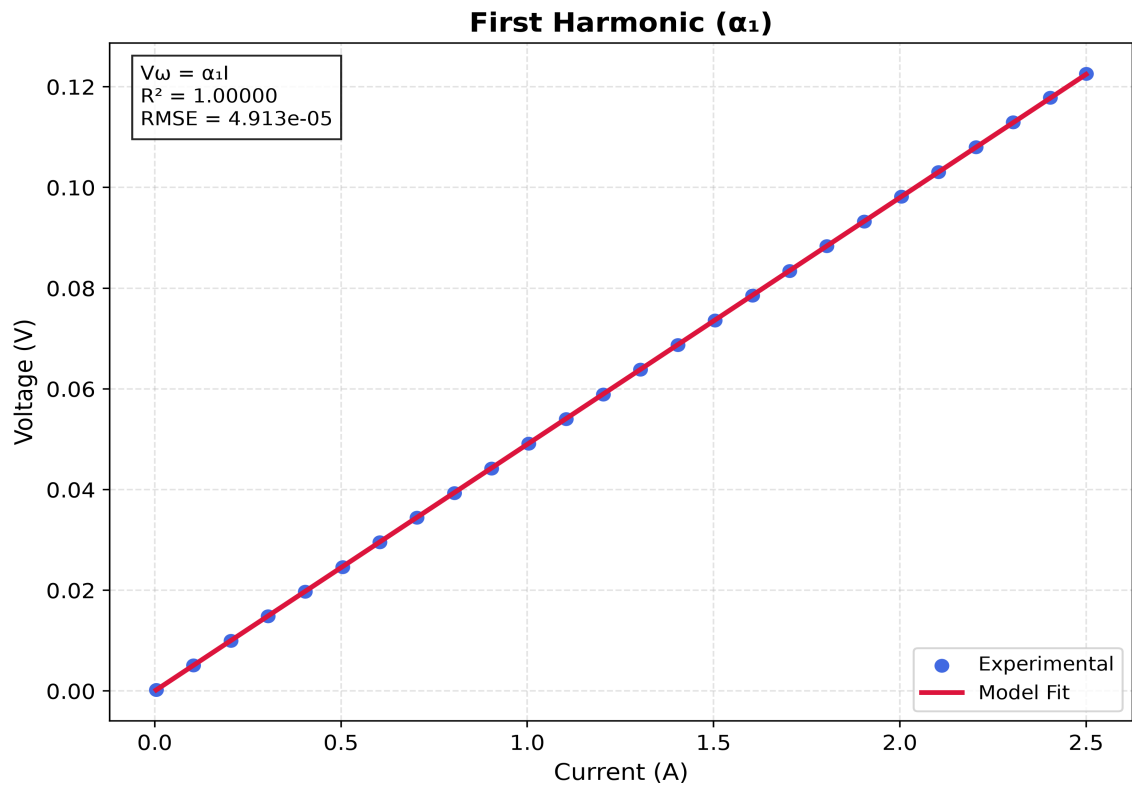
Generated Figures

Experimental Harmonic Data



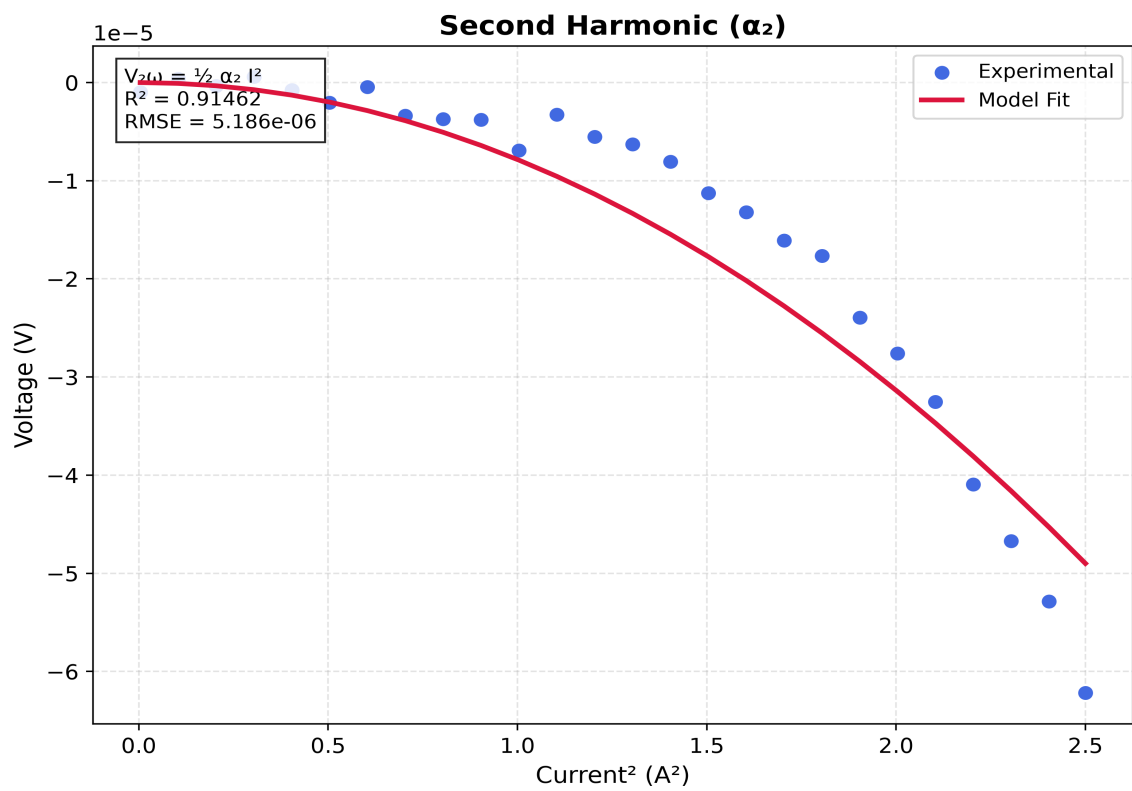
First Harmonic Fit (α_1)

Linear response obtained from the first harmonic experimental measurements.



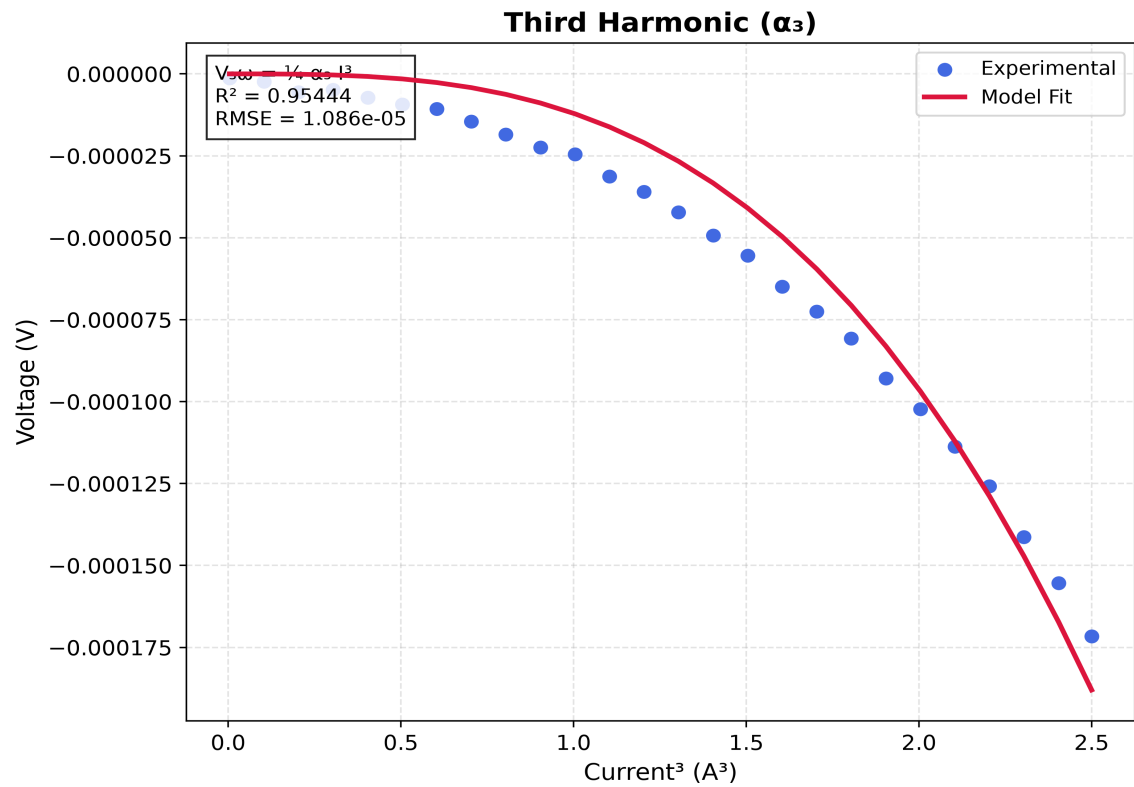
Second Harmonic Fit (α_2)

Quadratic nonlinear contribution extracted from the second harmonic measurements.



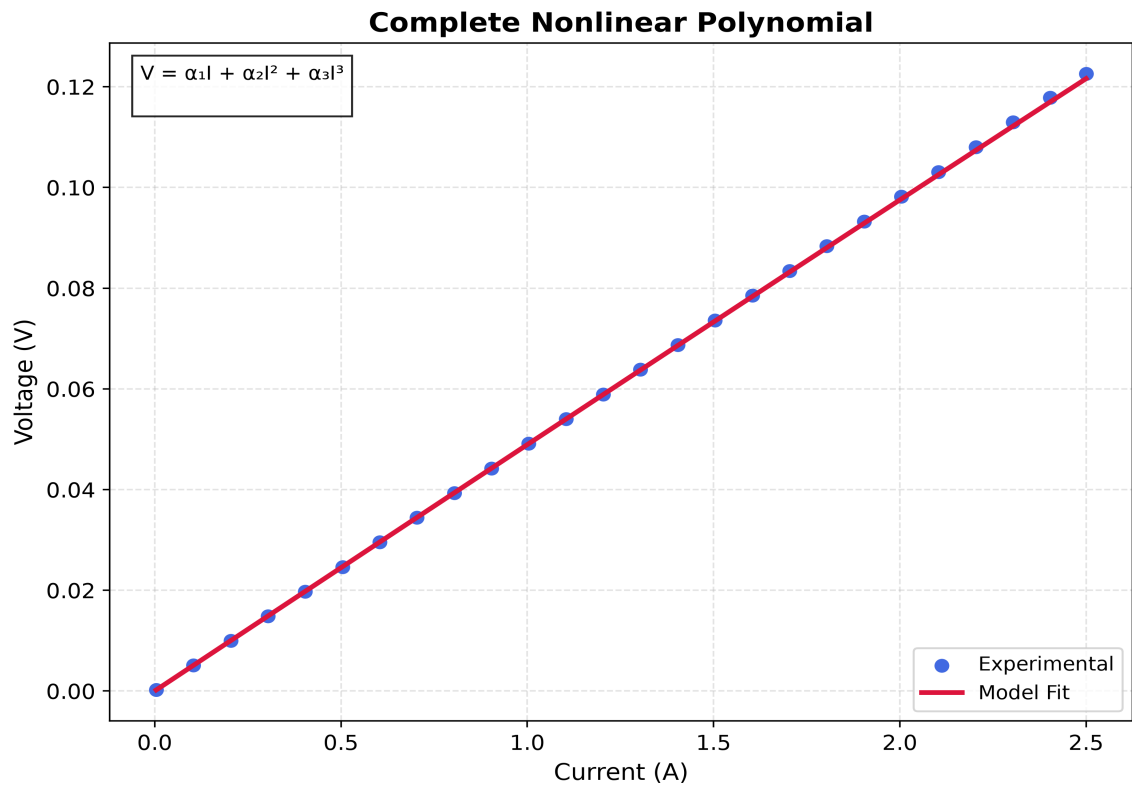
Third Harmonic Fit (α_3)

Third-order nonlinear response extracted from the third harmonic measurements.

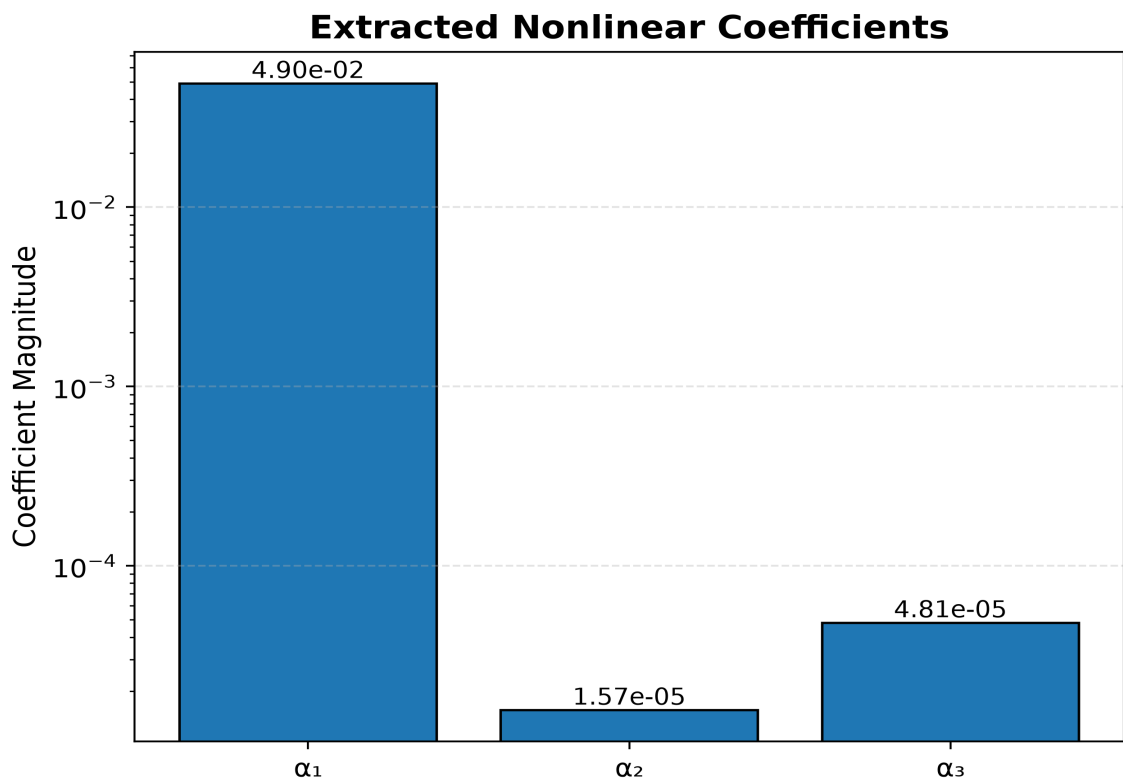


Complete Nonlinear Polynomial Model

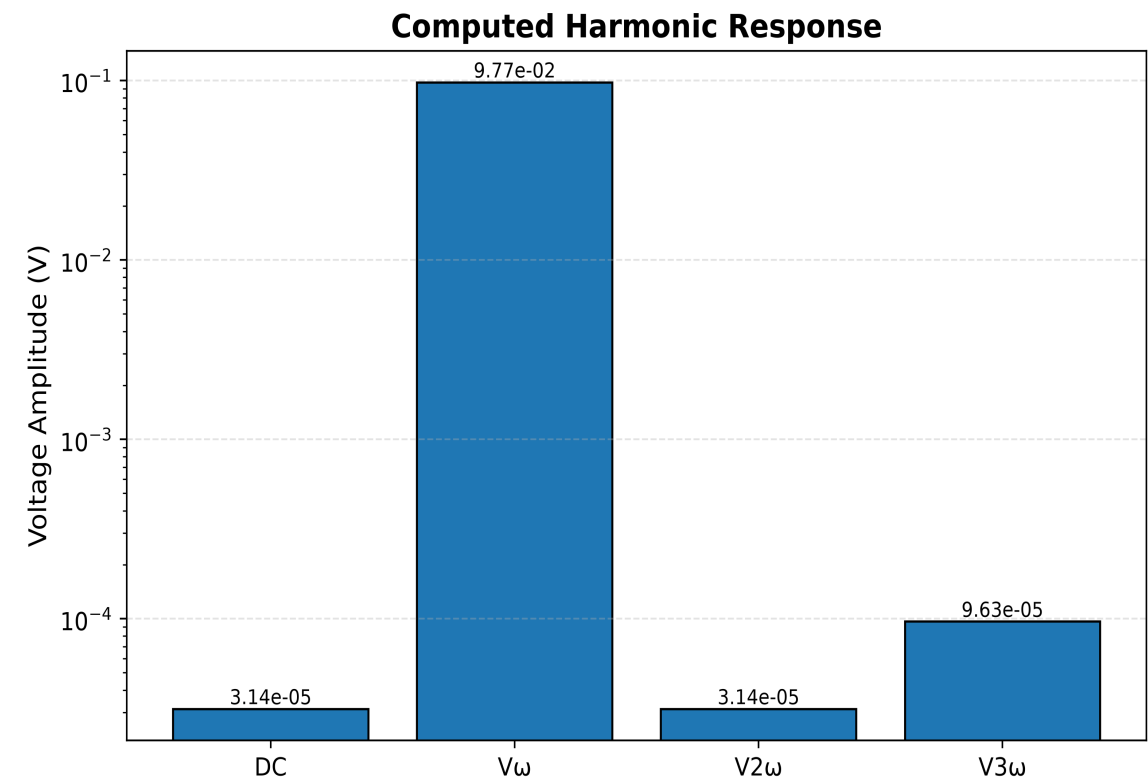
Prediction obtained using the complete nonlinear model $V = \alpha_1 I + \alpha_2 I^2 + \alpha_3 I^3$.



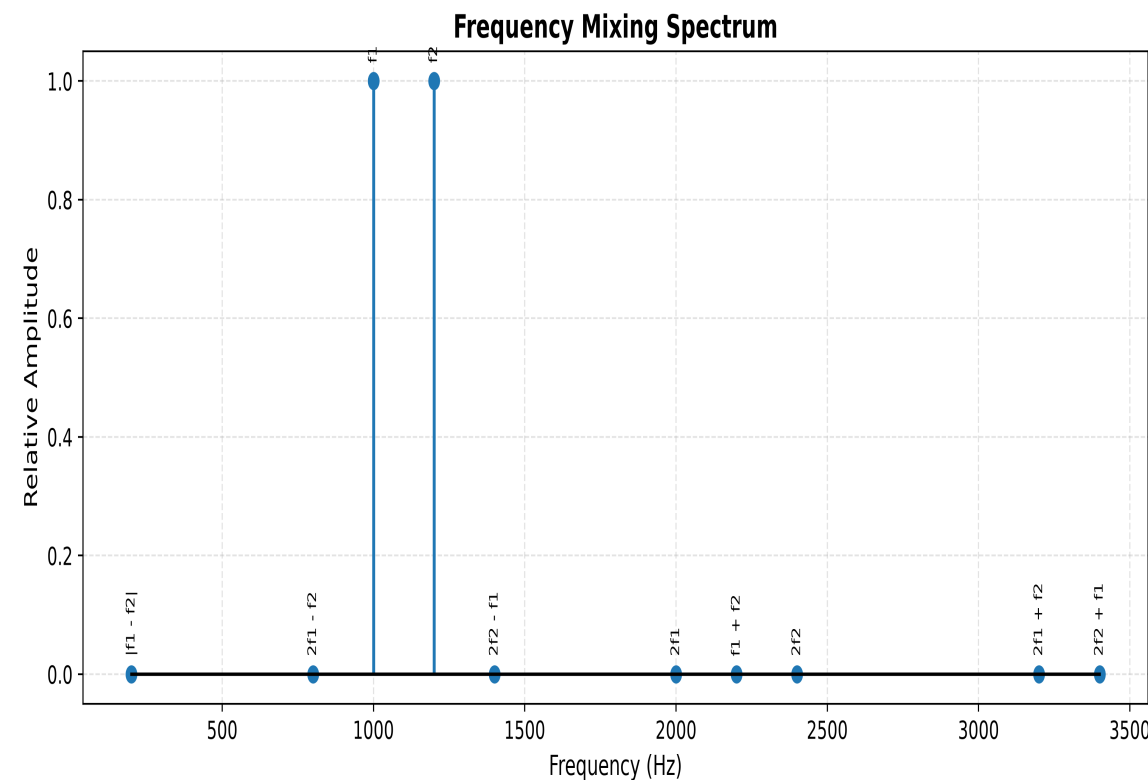
Extracted Nonlinear Coefficients



Computed Harmonic Response



Frequency Mixing Spectrum



Summary

The nonlinear coefficients extracted from the experimental measurements are $\alpha_1 = 4.898470\text{e-}02$, $\alpha_2 = -1.567957\text{e-}05$, and $\alpha_3 = -4.812896\text{e-}05$. Independent regression analyses were performed for the first, second and third harmonic datasets. The extracted coefficients were subsequently used to construct the nonlinear Hall Effect polynomial model, compute the harmonic voltage response and predict the frequency-mixing products generated under dual-frequency excitation. The excellent goodness-of-fit metrics indicate strong agreement between the theoretical model and the experimental data.